

RESEARCH ARTICLE

Influence of different functionality degree grafting monomers on polypropylene melt radical branching reaction and their structure

Chun Zhang^{1,2}  | Le Yang² | Li He² | Wei Gong² | Zhu Luo² | Wei Liu¹

¹School of Materials and Metallurgical Engineering, Guizhou Institute of Technology, Guiyang 550003, People's Republic of China

²National Engineering Research Center for Compounding and Modification of Polymer Materials, Guiyang 550014, People's Republic of China

Correspondence

Chun Zhang, School of Materials and Metallurgical Engineering, Guizhou Institute of Technology, Caiguan Road 1, Yunyan District, Guiyang 550003, Guizhou Province, People's Republic of China.

Email: zhangchun925@126.com

Long-chain-branched polypropylene (LCBPP) is one of polypropylenes (PPs) with high melt strength and good melt elasticity. Recently, due to its outstanding properties, LCBPP have been attracted increasingly attention in the field of development and characterization by the researchers all over the world. In this study, LCBPP was prepared by the melt radical branching reaction in a torque rheometer. The influences of various acrylate monomers with different functionality degrees on the structure and melt performance of PP products were investigated. The results indicated that grafting monomers with different functionality degrees made diverse influences on the branching density and branching chain length of branching PP products. With the increase of the functionality degree of grafting monomers, the branching level of PP products increased gradually and the "multiplicity" of branches became increasingly obvious. Besides, a higher reactivity of pentaerythritol triacrylate with hydroxyl than the similar molecular structured pentaerythritol tetraacrylate was confirmed. Furthermore, due to the high reactivity of dipentaerythritol penta(hexa)acrylate, branching and crosslinking reaction occurred simultaneously during the reaction process. As a result, the gel content increased and finally formed highly star branching structures with a shape of "dense and short."

KEYWORDS

degree of functionality, grafting monomer, long-chain branch (LCB), polypropylene (PP)

1 | INTRODUCTION

Polypropylene (PP) is one of the most widely used resins in the world relying on its many desirable and beneficial advantages, such as low density, high melting point, wide sources, superior mechanical properties, and chemical stability.¹ However, the application of the ordinary commercial polypropylene in some processes, eg, thermoforming, extrusion foaming, and blow molding, is limited because of its linear structure and low melt strength (MS). Thus, lots of researches are focusing on improving its MS.²⁻⁴

Some researches proposed that a small quantity of long-chain branches (LCBs) in PP could reinforce the MS obviously, which is owing to the augment of entanglements among the macromolecule chains.⁴⁻⁶ In addition, long-chain-branched polypropylene (LCBPP) exhibits obviously strain hardening behavior in the MS.^{7,8} The LCB can be introduced via copolymerized reaction during the stage of propene polymerization or through the branching modification of linear commercial PP using a method of post-reactive process. Due

to the inductive effect of methyl, the tertiary carbon atoms in PP own a great reactivity, which makes PP more vulnerable to be oxidized and thus then results in β -scission under the action of heat or high-energy rays. Therefore, the method of melt radical branching reaction, as a simple and effective way, is widely applied to prepare LCBPP.⁵⁻¹⁰ Nonetheless, this reaction process must be controlled strictly because of the complexity of free radical reaction, which mainly included coupled reaction among PP chains and degradation caused by β -scission.¹¹

Some grafting monomers are often added into the radical reaction system to stabilize the unstable tertiary carbon radicals. Those monomers are usually electron donors, such as styrene, acrylic amide, and acrylate. They can be combined with the PP tertiary carbon radicals and stabilize them via their conjugative effects. Moreover, the polyfunctional monomers (PFMs) can also act as a branching center and promote the branching reaction of PP chains.¹² However, PFM tends to self-polymerized via radical reaction initiated by initiators such as peroxide. It was proved that thiuram disulfide or

dithiocarbamate¹³⁻¹⁵ is capable to adjust radical reaction by reducing the instantaneous concentration of active radicals through reversible radical reaction, restrain the self-polymerization of the monomers and induce the formation of LCBs at the same time. Thus, one kind of LCBPP with high MS can be obtained.

In this work, various acrylates, with different degrees of functionality, were chosen as the grafting monomers for the branching reaction, and a torque rheometer was used as reactor. The influences of the degrees of functionality of grafting monomers on the melt radical branching reaction of PP and its branching structure were investigated, as well as the melt properties.

2 | EXPERIMENTAL

2.1 | Materials

Additive-free isotactic PP granule (S1003, MFR is 3.2 g/10 min) was supplied by Dushanzi Petrochemical Corporation, China. The antioxidant B215 was produced by SUNSHOW, China. The initiator, 2,5-dimethyl-2,5-di(tert-butylperoxy) hexane peroxide (DBHP), was obtained from AkzoNobel Polymer Chemistry Co Ltd, Ningbo, China. Neopentyl glycol diacrylate (NPGDA), trimethylolpropane triacrylate (TMPTA), pentaerythritol triacrylate (PETA), pentaerythritol tetraacrylate (PETEA), and dipentaerythritol penta(hexa)acrylate (DPP(H)A, DPPA/DPHA = 1:1) were all purchased from Jiangsu Hengyang Chemical Product Sales Department, China. The dithiocarbamate agent, zinc dimethyldithiocarbamate, was acquired from Hebi Uhoo Chemical Co Ltd, China. All reagents were used without further purification. The molecular structures of acrylate monomers are shown in Figure 1.

2.2 | Sample preparation

To make the additives disperse evenly, the corresponding amount of acrylate monomers, DBHP, Irganox B215, and zinc dimethyldithiocarbamate were firstly dissolved in 10-mL acetone and then sprinkled in the PP pellets and followed by stirring for 5 minutes. After the volatilization of acetone in a ventilating cabinet at room temperature, the mixture was transferred in a torque rheometer (Shanghai, China) at 180°C and 80 rpm for a certain time to obtain modified PP. Finally, the samples were smashed to granules in a pulveriser. The contents of all formulations in this study are listed in Table 1.

2.3 | Characterization

2.3.1 | Purification of sample

The reacted PP samples were dissolved in hot xylene at 138°C, and then the solutions were charged into acetone at room temperature. The PP and PP-g-acrylate were precipitated out, while the unreacted acrylate monomer and homopolymerized acrylate remained soluble. The PP and PP-g-acrylate were separated by filtration and then was dried at 80°C under vacuum for 24 hours.¹⁶ The dried purified sample was used for the subsequent characterization.

2.3.2 | Gel determination

About 0.3 g modified PP sample was packed with phosphor bronze net of 400 meshes and then extracted in boiling xylene for 24 hours in a Soxhlet extractor. After extraction, the bronze package was dried at 80°C for at least 24 hours. The weights of PP sample, bronze net, and the dried bronze package were marked as w_0 , w_1 , and w_2 , respectively. The gel content could be estimated by the following formula:

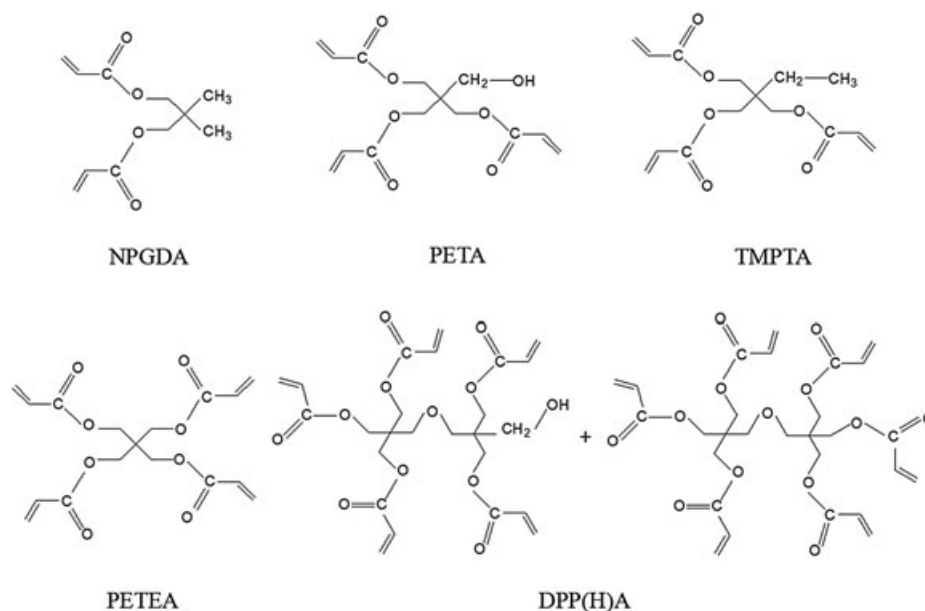


FIGURE 1 Structures of 5 kinds of acrylates with different degrees of functionality. DPP(H)A, dipentaerythritol penta(hexa)acrylate; NPGDA, neopentyl glycol diacrylate; PETA, pentaerythritol triacrylate; PETEA, pentaerythritol tetraacrylate; TMPTA, trimethylolpropane triacrylate

TABLE 1 The detailed composition of different modified polypropylene (PP) samples (unit: g)

Sample	PP	NPGDA(2) ^a	TMPTA(3)	PETA(3)	PETEA(4)	DPP(H)A(5/6)	DBHP	ZDMC	B215
PP ₀	100	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
PP _{DBHP}	100	N/A	N/A	N/A	N/A	N/A	0.1	0.1	0.2
PP _{NPGDA}	100	1.2	N/A	N/A	N/A	N/A	0.1	0.1	0.2
PP _{TMPTA}	100	N/A	1.2	N/A	N/A	N/A	0.1	0.1	0.2
PP _{PETA}	100	N/A	N/A	1.2	N/A	N/A	0.1	0.1	0.2
PP _{PETEA}	100	N/A	N/A	N/A	1.2	N/A	0.1	0.1	0.2
PP _{DPP(H)A}	100	N/A	N/A	N/A	N/A	1.2	0.1	0.1	0.2

Abbreviations: DBHP, 2,5-dimethyl-2,5-di(tert-butylperoxy) hexane peroxide; DPP(H)A, dipentaerythritol penta(hexa)acrylate; NPGDA, neopentyl glycol diacrylate; PETA, pentaerythritol triacrylate; PETEA, pentaerythritol tetraacrylate; TMPTA, trimethylolpropane triacrylate; ZDMC, zinc dimethyldithiocarbamate.

^aThe number in the parenthesis represents the functionality degree of the monomer.

$$\text{Gel} = \frac{w_2 - w_1}{w_0} \times 100\%. \quad (1)$$

All the tested specimens were sampled at the time of second torque peak in the torque-time curves.

2.3.3 | Fourier transform infrared spectroscopy

Fourier transform infrared (FTIR), Nexus670, (Nicolet, USA) was used to characterize pristine PP and modified PP samples. The purified sample was pressed into thin slice with spectrum pure potassium bromide (KBr, SP) at room temperature and 14 MPa for the measurement.

2.3.4 | Solid-state ¹³C nuclear magnetic resonance spectrum

Solid-state ¹³C nuclear magnetic resonance (NMR) experiments were performed on a Bruker 400 M/600 M spectrometer. All ¹³C NMR experiments were conducted at room temperature, 20-Hz spinning rate, 90° pulse angle, 5-second acquisition time, and 10-second relaxation delay.

2.3.5 | Rheological measurements

A rotational rheometer (ARES-G2, TA Instruments, USA) with a parallel plate (25 mm in diameter) was used to characterize rheological behavior of PP samples. The dried purified samples were compressed into pallets, with a thickness of 2.0 mm and a diameter of 25 mm. The small amplitude oscillatory shear was performed in the frequency range of 0.03 to 100 rad/s at 190°C. The applied strain was 1% so as to ensure all samples could maintain in the linear viscoelastic regime.

The melt extension was performed using a capillary rheometer (Rosand RH7, Malvern) for Haul-off test at 190 ± 0.5°C, equipped with a flat entrance angle (180°) of diameter of 2.0 mm and an aspect ratio L/D = 20.

2.3.6 | High-temperature gel permeation chromatography

A high-temperature gel permeation chromatography (HTGPC, PL-220, UK), equipped with a triple detection, was applied to obtain the molecular weight and molecular weight distribution of pristine PP and modified ones. Before testing, the samples were dissolved in 1,2,4-trichlorobenzene with 1.25 w/v% butylated hydroxytoluene as antioxidant at 150°C and wobbled for 3 hours on a shaker (PL-SP 260VS) and then leached with a mesh filter gun to eliminate the insoluble gels in the samples. All the measurements were taken place at a temperature of 150°C.

3 | RESULTS AND DISCUSSION

3.1 | The torque curves of PP's melt radical branching reaction

Figure 2 shows the torque curves of pristine PP (PP₀) and modified PP samples with different acrylate monomers. It is obvious that there are 2 peaks in the curves for all the modified PP samples. With the increase of monomers' functionality degree, the reactive possibility between PP tertiary carbon radicals and grafted monomers increase, and the reaction progress is also accelerated at the same time, thus the branching reaction even crosslinking among PP chains could take place in a great extent. As a result, a higher-equilibrium torque is observed and the onset of the second peak increasingly becomes earlier. The gel contents of PP_{NPGDA}, PP_{TMPTA}, PP_{PETA}, PP_{PETEA}, and

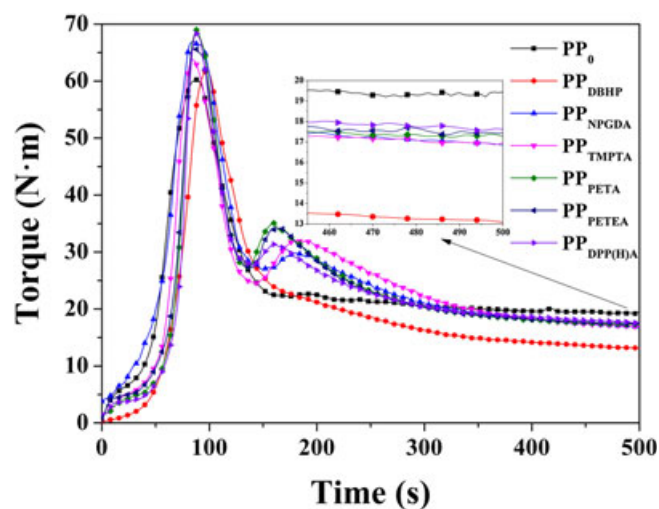


FIGURE 2 Torque curves of virgin polypropylene (PP) and modified PP with different monomers. DBHP, 2,5-dimethyl-2,5-di(tert-butylperoxy) hexane peroxide; DPP(H)A, dipentaerythritol penta(hexa)acrylate; NPGDA, neopentyl glycol diacrylate; PETA, pentaerythritol triacrylate; PETEA, pentaerythritol tetraacrylate; TMPTA, trimethylolpropane triacrylate [Colour figure can be viewed at wileyonlinelibrary.com]

PP_{DPP(H)A} are 0.1%, 0.4%, 1.2%, 1.8%, and 7.8%, respectively. For PP_{DPP(H)A}, the gel content exceeds 5%, which indicates a certain degree of crosslinking. As for the tri-functionality monomers (TMPTA and PETA), the hydroxyl (–OH) in PETA increases the polarity of the monomer molecule, which suggests a higher reaction activity of PETA than that of TMPTA. Meanwhile, PP_{PETA} shows an earlier and higher second peak than that of PP_{TMPTA} (see Figure 2). In addition, compared with the grafted samples, PP_{DBHP} exhibited only 1 torque peak (due to the melting of PP) and lower equilibrium torque value (13.10 N·m), indicating that apparent degradation rather than branching reaction occurs.

3.2 | The FTIR spectroscopy

The FTIR spectra of the purified specimens are shown in Figure 3A. For all the modified samples, bands at about 1743 cm^{−1} corresponded to the stretching vibration of the carbonyl group of the ester in the PFMs¹⁷ which is absent in pristine PP could be obviously observed, indicating that the PFMs were grafted onto the PP backbone. According to the normalized FTIR spectra, the relative grafting degree (RGD) could be calculated as follows⁸:

$$\text{RGD} = \frac{A_{1743}}{nA_{1169}} \times 100\%, \quad (2)$$

where n is the degree of functionality (double bond number) of the PFM; A_{1743} is the absorbance at 1743 cm^{−1}, characterizing the carbonyl groups of the ester in the monomer; and A_{1169} is the absorbance at 1169 cm^{−1}, characterizing the methyl groups on the PP backbone. The RGD values of different polyfunctional monomers for PP_{NPGDA}, PP_{TMPTA}, PP_{PETA}, PP_{PETEA}, and PP_{DPP(H)A} are 0.09, 0.09, 0.10, 0.10, and 0.10, respectively, indicating that the RGD is nearly independent to the degree of functionality of the monomer.

3.3 | Solid-state ¹³C NMR spectrum

The structures of the modified samples are investigated in detail using solid state ¹³C NMR. The typical spectra (PP₀ and PP_{PETA}) are presented in Figure 3B,C. For PP₀, the typical resonances at 21.47, 26.05, and 44.11 ppm respectively correspond to the primary carbon

(–CH₃), tertiary carbon (–CH), and secondary carbon (–CH₂) on the PP backbone. For the modified sample PP_{PETA}, an apparent new resonance at 32.42 ppm could be observed which is due to the quaternary carbon (marked as “br” in Figure 3C) in the monomer PETA, thus further verifying that the acrylate monomer is grafted onto PP backbone. Moreover, resonances in the range of 100 to 150 ppm for double bond are not seen in the ¹³C NMR spectra of all modified samples, indicating that all double bonds in these acrylate monomers are reacted with the PP chains and star branched PP is formed. The grafting percentages (GPs) of the monomer could be calculated according to $\text{GP} = nI_{\text{br}} / I_{\text{total}}$, where n is the degree of functionality of the PFM, I_{br} , and I_{total} are the integral areas of quaternary carbon (branch point) resonance and total intensity of carbons, respectively. The GP values are summarized in Table 2. For the samples modified by polyfunctional acrylate monomers, the GP increases with the increasing of functionality of the monomers, indicating that the functionality degree of monomers acts a very important role in the occurrence of branching reactions.

3.4 | The rheological properties of PP samples

Rheological properties of polymers in the linear viscoelastic region are related to the molecule topological structure. When the PP chains transform from a linear molecule to a branched or cross-linked structure, the storage modulus (G') and loss angle (δ) at low-frequency

TABLE 2 Characterization results of pristine polypropylene (PP) and modified PP samples

Sample	MFR ^a (g/10 min)	GP	M_w (g/mol)	M_w/M_n	g
PP ₀	3.20	0	363 180	6.04	1.00
PP _{NPGDA}	6.28	0.097	245 536	6.78	0.71
PP _{PETA}	0.40	0.215	301 965	6.44	0.62
PP _{PETEA}	0.54	0.207	328 968	6.43	0.64
PP _{DPP(H)A}	0.29	0.226	325 909	6.25	0.59

Abbreviations: DPP(H)A, dipentaerythritol penta(hexa)acrylate; GP, grafting percentage; MFR, melt flow rate; NPGDA, neopentyl glycol diacrylate; PETA, pentaerythritol triacrylate; PETEA, pentaerythritol tetraacrylate.

^aMeasured at 230°C, 2.16 kg.

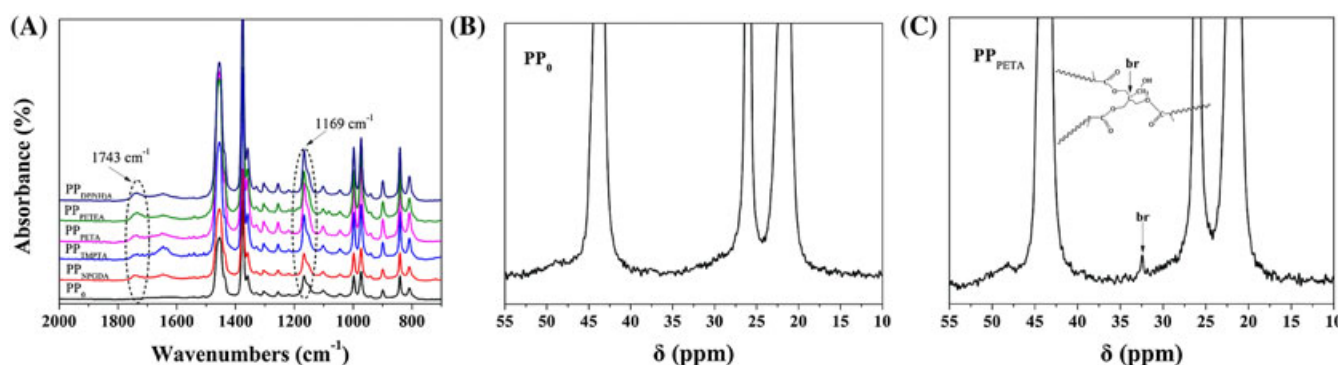


FIGURE 3 A, Normalized Fourier transform infrared spectra of pristine PP and modified PPs after purification. B,C ¹³C nuclear magnetic resonance (NMR) spectra of pristine PP and PP_{PETA} at room temperature. DPP(H)A, dipentaerythritol penta(hexa)acrylate; NPGDA, neopentyl glycol diacrylate; PETA, pentaerythritol triacrylate; PETEA, pentaerythritol tetraacrylate; TMPTA, trimethylolpropane triacrylate [Colour figure can be viewed at wileyonlinelibrary.com]

region will increase due to the augment of the entanglements between the PP chains.¹⁸ Figure 4A,B respectively shows the storage modulus (G') and the complex viscosity (η^*) versus angular frequency (ω) of pristine PP and modified PP samples at 190°C. Both of G' and η^* of modified PPs at low-frequency region increase obviously compared with those of pristine PP, especially for PP_{PETA} and PP_{DPP(H)A}. The presence of hydroxyl (–OH) in PETA increases the molecule polarity and enhances its reaction activity, thus PP_{PETA} exhibits apparent branching characteristic (with some slight crosslinking structure). Due to the double bonds in DPP(H)A as well as hydroxyls in DPPA, both branching and crosslinking reaction took place during the melt radical reaction and some gels generate at the same time, thus leading to the highest values in G' and η^* of PP_{DPP(H)A} at the low-frequency region. These experiment results are consistent with those acquired from the torque curves. It is reported that Cole-Cole plot (η'' – η'), Figure 4C could be used to illustrate the long relaxation process.¹⁹ As shown in Figure 4C, the distinctions between pristine PP and modified PPs are obvious. For the pristine PP, the Cole-Cole plot exhibits a shape of similar semicircle owing to its linear molecule structure. However, all the Cole-Cole plots of PPs modified by the acrylate monomers deviate from semicircle and show an uptrend at high viscosity, indicating that a different relaxation mechanism and a

long relaxation time.¹⁹ Besides, the curve of PP_{NPGDA} is lower than that of pristine PP, which indicates that degradation reaction plays a main role in this free radical reaction due to the poor stabilization effect of NPGDA on the PP tertiary carbon radical.¹⁶ When adding DBHP alone, both G' and η^* of PP_{DBHP} decrease and the Cole-Cole plot exhibits a shape of minor radius semicircle compared with PP₀, which reveals that DBHP can mainly cause the degradation rather than branching reaction when the PFM is absent.

Extensional flow experiment is often used to reflect the extensional behavior of polymers. When LCB exists, PP melt can display more obvious strain-hardening behavior. It is known from literatures that the extensional flow data can be fitted using parameters determined solely by the shear measurements.^{20,21} Figure 4D shows the tensile stress growth coefficients η^+ of the melt of pristine PP and modified PP samples at a strain rate of 1.0 s^{−1}. As shown, η^+ of all samples increase gradually with the increasing run time, while compared with pristine PP, the modified samples exhibit more obvious strain-hardening behavior, especially for PP_{DPP(H)A}. That further confirms the introduction of LCB structures in the PP chains via the melt reaction processing.

In the Haul-off test curve, the “melt strength” is defined as the maximum force before the melt strand breaking. When LCB exists,

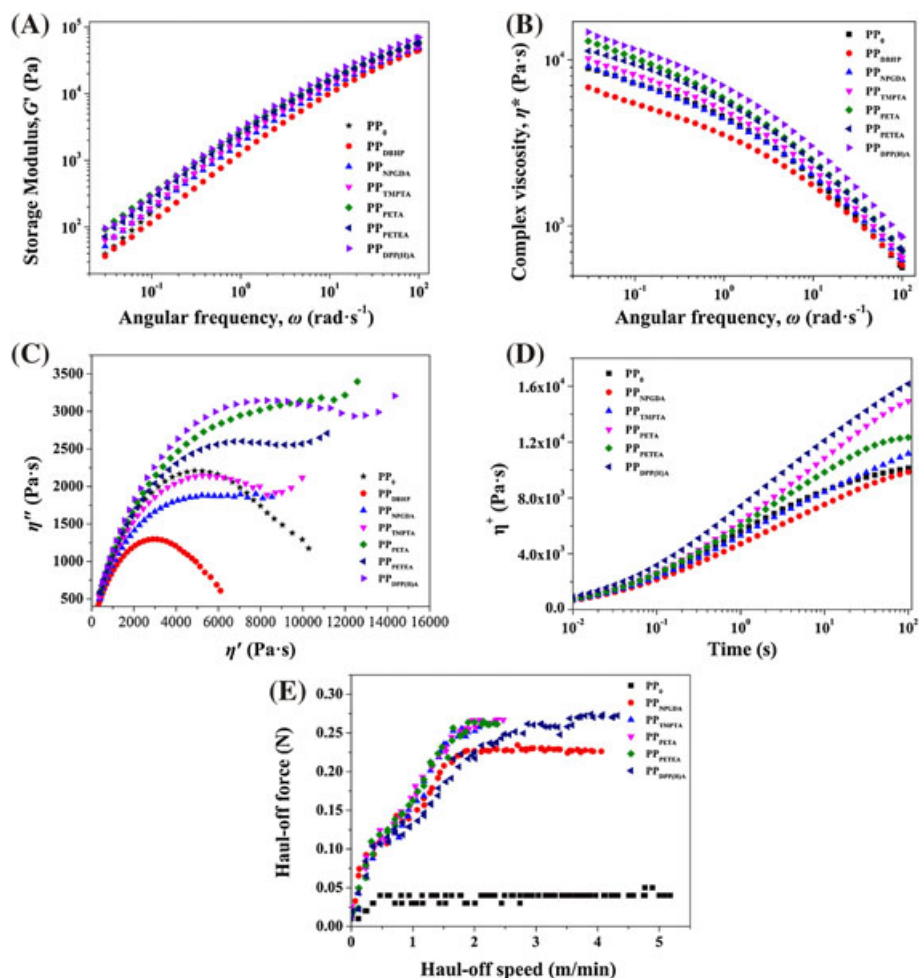


FIGURE 4 A, Storage modulus (G') versus angular frequency (ω) for pristine PP and modified PPs at 190°C. B, Complex viscosity (η^*) versus angular frequency (ω) for pristine PP and modified PPs at 190°C. C, Cole-Cole plot of the pristine PP and modified PPs at 190°C. D, Tensile stress growth coefficients η^+ of PP samples at a strain rate of 1.0 s^{−1}. E, Haul-off force vs haul-off speed of PP₀ and modified samples at 190°C [Colour figure can be viewed at wileyonlinelibrary.com]

polymer often exhibits high MS. For pristine PP and modified PP samples, the test results are clearly observed from Figure 4E. For pristine PP, the MS is just about 0.04 N, and in the experimental test range, it does not break. While those of the samples modified by polyfunctional monomers are much higher than that of pristine PP, and the MS increases with the increasing of functionality of the monomers, indicating obvious strain hardening for the modified PP when LCB is introduced.

3.5 | The analysis of PP branching chains

3.5.1 | Characterization of the branch structure

Gel permeation chromatography is considered as one of the efficient ways to characterize the LCBs in polymers. Here, the molecule weight (M_w) and intrinsic viscosity ($[\eta]$) of every elution fractions are measured by a HTGPC equipped with a viscosity detector. The molecular weight (M_w), molecular weight distribution (MWD, M_w/M_n), and melt flow rate (MFR) of PP samples are listed in Table 2. It is found that the PP samples modified by the melting radical reaction initiated by DBHP exhibit a certain degree of decrease in molecule weight and the MWD become slightly wider comparing with pristine PP. Moreover, apparent decreases in MFR can be seen when the functionality degree of monomers exceeds 3, indicating high chain entanglement among the LCB chains.

The following Mark-Houwink equation is used to study the branching structure of samples²²:

$$[\eta] = KM_w^\alpha \quad (3)$$

where K and α are 2 constants. Generally, a linear relation between the $[\eta]$ and its M_w can be presented in the log-log coordinate for a linear polymer, while a deviation from the linearity can be observed in long-chain-branched polymers.²³

Figure 5A shows that the $[\eta]$ value of PP samples is as a function of their respective M_w in log-log coordinate. For the pristine PP (PP₀), a visualized linear relationship is presented because of the linear molecule structure. However, a certain degree of deviation from the linearity exists in all the modified PPs, which indicates that the long branching chains are grafted on the PP skeleton. Besides, a larger degree of deviation suggests a higher branching degree.²⁴ In addition, NPGDA also affects on the branching of PP, and the $[\eta]$ - M_w curve of PP_{NPGDA} displays a low degree of deviation from the linearity indicating a low degree of branching. Enhancing monomer's degree of functionality benefits the branching reaction of PP. However, DPP(H)A could lead to a high branching level and exacerbate the crosslinking reaction at the same time, which results in an increase of gel content (7.8%) for PP_{DPP(H)A}. Moreover, a higher branching degree of PP_{PETA} than that of PP_{PETEA} is observed, which is ascribed to that the hydroxyl in PETA increased the polarity of molecule and enhanced the reactivity than nonpolar PETEA. The gel content of PP_{PETA} (1.2%) is relatively low, indicating that PETA is suitable for the branching modification of PP.

To further verify the branching structure of PP, the branching density of LCB (the average number of branches per PP chain) is calculated.¹² A common model of Zimm and Stockmeyer is applied to reflect the branching structure of the modified samples.²⁴ The ratio of the mean square radii of gyration, g , of the branching and the linear

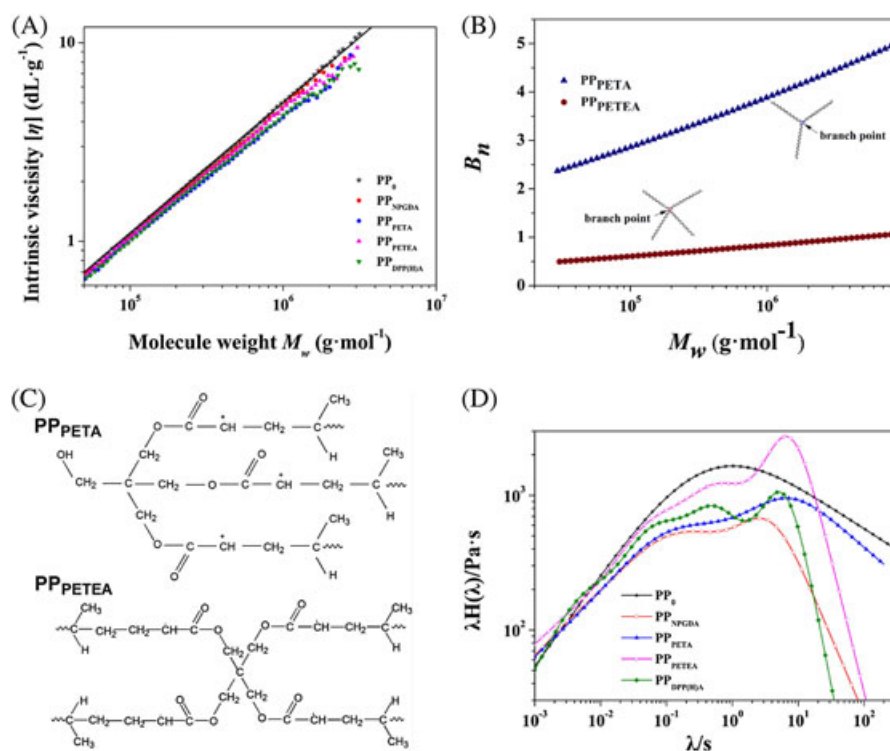


FIGURE 5 A, $[\eta]$ - M_w plots of the pristine polypropylene (PP) and modified PPs in log-log coordinate at 150°C. B, Branching density B_n of modified PP. C, PP star chains formed via pentaerythritol triacrylate (PETA) and pentaerythritol tetraacrylate (PETEA) as branching center. D, Weighted relaxation spectrum for the pristine PP and modified PPs at 190°C. DPP(H)A, dipentaerythritol penta(hexa)acrylate; NPGDA, neopentyl glycol diacrylate [Colour figure can be viewed at wileyonlinelibrary.com]

PP could be obtained from their intrinsic viscosities ($[\eta_B]$ and $[\eta_L]$), with the following relationship, and a typical value of the exponent ε is 0.75 for star branching or random branching. The average g value of modified PP samples are calculated in Table 2. It can be found that all the values of g of the modified PP are smaller than 1 and decreased with the increasing of functionality of the monomers.

$$g = ([\eta_B]/[\eta_L])^{1/\varepsilon} \quad (4)$$

Moreover, it is noted that the value of g must be calculated with the same molecule weight, so that a proper molecule weight range needs to be adopted. Herein, PP_{PETA} and PP_{PETEA} are taken for example, and it is assumed that each double bond in reacted monomers is coupled with 1 PP chain, as a result, 3- or 4-arm star branching chains are formed (shown as Figure 5C).

According to the following equations,²³ the average number of branches per PP chain, B_n , can be estimated. The calculated results are shown in Figure 5B.

$$\text{For 3-arm branching: } g = (1 + B_n/7)^{1/2} + 4B_n/9\pi \quad (5)$$

$$\text{For 4-arm branching: } g = [(1 + B_n/6)^{1/2} + 4B_n/3\pi]^{-1/2} \quad (6)$$

It is apparently observed that the B_n values of PPPETEA (in the range of 0.5 to 1.0) is lower than that of PPPETA, which coincides well with the $[\eta]$ - M_w results. PPPETA's B_n values in the part of high molecular weight exceeds 5.0, which indicates a high branching degree. In addition, the B_n values of PPPETA and PPPETEA are dependent on the molecular weight, higher B_n values of the long-chain molecule could be seen in the high molecular weight region, and especially in PPPETA, a high branching degree is confirmed.

The branching structure of modified PP is verified by the HTGPC results at the chemical structure level, but the length of the LCBs could not be reflected by above experiments.

3.5.2 | Weighted relaxation spectrum and the calculation of LCB length

As mentioned above, Cole-Cole plot ($\eta''-\eta'$) is used to illustrate the long relaxation process (see Figure 4C). However, the entire relaxation behaviors of PP specimens are hardly reflected only depended upon the Cole-Cole curves, due to the restriction of observation time. To further analyze the molecule structure of branching PP and make it more obvious, the weighted relaxation spectra of pristine PP and modified PP in a wider range, which suggests an average level of the relaxation time for the whole molecular chains in this system, are calculated from dynamic modulus (G' and G'') data according to the following relations using a GENEREG program^{16,25}:

$$G'(\omega) = \int_{-\infty}^{\infty} \frac{H(\lambda)\omega^2\lambda^2}{1 + \omega^2\lambda^2} d\ln\lambda, \quad G''(\omega) = \int_{-\infty}^{\infty} \frac{H(\lambda)\omega\lambda}{1 + \omega^2\lambda^2} d\ln\lambda, \quad (7)$$

where $H(\lambda)$ is the continuous relaxation spectrum, ω is the sweep frequency, and λ is the relaxation time of PP chains.

The calculated weighted relaxation spectra are shown in Figure 5D. For the pristine PP, only 1 characteristic relaxation time can be confirmed, which relates to the relaxation process of linear

molecules. For the LCBPP, the relaxation time peak is wider than pristine PP, and the peak value shifts to the region of long-time relaxation. Besides, multiple relaxation time peaks appear in the region of long-time relaxation, and it is considered that this slower relaxation behavior is corresponding to the LCBPP molecule. With the increase of the functionality degree of grafted monomer, relaxation time of PP_{NPGDA}, PP_{PETA}, and PP_{PETEA} become longer gradually. The weighted relaxation spectra of PP_{NPGDA} and PP_{PETA} display double relaxation time peaks, which are respectively corresponding to the relaxation of linear molecule and branching molecule. By contrast, PP_{PETA} owns longer relaxation time and wider distribution than those of PP_{NPGDA}. Triple relaxation time peaks and quadruple relaxation time peaks are respectively exhibited in the weighted relaxation spectra of PP_{PETEA} and PP_{DPP(H)A}, but the relaxation time of PP_{DPP(H)A} is relatively shorter than those of PP_{PETA} and PP_{PETEA}.

It was reported by Watanabe that a relationship for viscoelastic linear polymer could be expressed as²⁶

$$\tau_L = 3\tau^*(a)N^3(1 - 1.47N^{-0.5})^2, \quad (8)$$

where τ_L is the characteristic relaxation time of linear chains, $\tau^*(a)$ is the internal equilibration time of each entangle segment, $N = M / M_e$ is the segment number of a linear chain, and M_e (≈ 5600 g/mol for PP) is the molecular mass between entanglements. Depending on the HTGPC results, the M_w value of pristine PP (PP₀) was 363180 g/mol, and the N value (64.9) of linear pristine PP could be calculated.

The relaxation time of long-branched polymers are estimated by the Ball-McLeish model²⁷:

$$\tau_B = \tau^*(a) \exp(vN_B/3), \quad (9)$$

where τ_B is the characteristic relaxation time of branching chains, N_B is the entanglement segment number in branching chain ($N_B = M_B / M_e$), and v is a constant ($v = 15/8$). Thus, the length of the branched chain N_B and M_B can be calculated through the characteristic relaxation time ratio between linear chain and branched chain:

$$\frac{\tau_B}{\tau_L} = \frac{\exp(vN_B/3)}{3N^3(1 - 1.47N^{-0.5})^2}; \quad (10)$$

τ_L and τ_B can be obtained from the weighted relaxation spectra. The characteristic relaxation times, branch length, and branch molecular weight are listed in Tables 3 and 4.

As shown in Table 3, modified branching PPs exhibit multiple long-chain branching structure with unequal branch lengths corresponding to the multiple relaxation times in the weighted relaxation time

TABLE 3 The characteristic relaxation time of pristine polypropylene (PP) and modified PP samples

Sample	τ_L , s	τ_{B1} , s	τ_{B2} , s	τ_{B3} , s	τ_{B4} , s
PP ₀	1.029	N/A	N/A	N/A	N/A
PP _{NPGDA}	1.029	0.254	2.617	N/A	N/A
PP _{PETA}	1.029	0.329	6.335	N/A	N/A
PP _{PETEA}	1.029	0.053	0.680	6.435	N/A
PP _{DPP(H)A}	1.029	0.005	0.070	0.488	5.075

TABLE 4 The length and molecule weight of polypropylene (PP) branches (unit: g/mol)

Sample	N	N_{B1}	N_{B2}	N_{B3}	N_{B4}	M_{B1}	M_{B2}	M_{B3}	M_{B4}
PP ₀	64.9	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
PP _{NPGDA}	64.9	18.9	22.6	N/A	N/A	105 840	126 560	N/A	N/A
PP _{PETA}	64.9	19.3	24.1	N/A	N/A	108 080	134 960	N/A	N/A
PP _{PETEA}	64.9	16.4	20.5	24.1	N/A	91 840	114 800	134 960	N/A
PP _{DPP(H)A}	64.9	12.6	16.8	20.0	23.7	70 560	94 080	112 000	132 720

Abbreviations: DBHP, 2,5-dimethyl-2,5-di(tert-butylperoxy) hexane peroxide; DPP(H)A, dipentaerythritol penta(hexa)acrylate; NPGDA, neopentyl glycol diacrylate; PETA, pentaerythritol triacrylate; PETEA, pentaerythritol tetraacrylate; TMPTA, trimethylolpropane triacrylate; ZDMC, zinc dimethyldithiocarbamate.

spectra. With the increasing of the functionality degree of grafted monomers, the multiplicity of branches becomes increasingly obvious. Two kinds of branching structures with different length can be confirmed in both PP_{NPGDA} and PP_{PETA}, and the branch length of PP_{NPGDA} is shorter than that of PP_{PETA}. The entanglement segment numbers of the 2 kinds of branching chains in PP_{PETA} are respectively 19.3 and 24.1 (shown in Table 4), about one-thirds of the linear PP main chain. There are 3 characteristic relaxation peaks of PP_{PETEA} in the weighted relaxation time spectrum, and it is correspondingly observed 3 kinds of different length branches. The “multiformity” in the degree of functionality of grafted monomers made the “multiplicity” of branching PP chains. In the $[\eta]$ - M_w plot (Figure 5A) of PP_{DPP(H)A}, it displays high degree of deviation from linearity, indicating apparent branching structure. From the point of view of branching chain length, the quadruple relaxation peaks of PP_{DPP(H)A} in its weighted relaxation time spectrum reveal that it owns 4 kinds of branching chains with different lengths. However, the relaxation times of PP_{DPP(H)A} are relatively shorter than those of PP_{PETA} and PP_{PETEA}, and shorter branching chains (N_{B1} and N_{B2}) appear correspondingly. It is manifested that the polyfunctionality (5 or 6) of DPP(H)A is able to provide more branch points than other monomers to promote the branching reaction of PP, while high reactivity certainly aggravate crosslinking reaction (the gel content of PP_{DPP(H)A} is 7.8%), and the part of gels generated from the crosslinking reaction cannot couple with radicals in the system. Thus, the unstable radicals can lead the degradation of PP chains, and one kind of relatively “dense and short” branches can be obtained.

4 | CONCLUSIONS

Long-chain branched PP was prepared in a torque rheometer, and the existence of LCBs was verified via FTIR, ^{13}C NMR, HTGPC, and rheological measurement results. Grafting monomers with different degrees of functionality made diverse influences on the branching density and branching chain length of PP branching product. With the increase in the functionality degree of grafting monomer, the branching level of PP increased gradually and the “multiplicity” of branches became increasingly obvious. Pentaerythritol triacrylate, with hydroxyl functional group, showed a higher reactivity than that of PETEA. For the high functionality degree monomer DPP(H)A, branching reaction and crosslinking of PP took place simultaneously under the action of DPP(H)A because of its so high reactivity, as well

as the gel content increased. And finally, a relatively “dense and short” branching structure of formed PP was confirmed.

In respect of branching chain molecular weight, the branch molecular weights of modified PPs (M_B , as shown in Table 4) were all larger than $2.5M_e$, conformed with the condition of LCB, which indicated that the architecture of PP transformed from a linear structure into a long branching structure after the melt radical branching modification.

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ORCID

Chun Zhang  <http://orcid.org/0000-0003-3038-6269>

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